

- Carefully weigh the risks and benefits of pharmacological treatment against the risks and benefits of nontreatment with antidepressants and make sure to document this in the patient's chart
- Monitor patients face-to-face regularly, particularly during the first several weeks of treatment
- Use with caution, observing for activation of known or unknown bipolar disorder and/or suicidal ideation, and inform parents or guardians of this risk so they can help observe child or adolescent patients

Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- Not generally recommended for use during pregnancy, especially during first trimester
- Nonetheless, continuous treatment during pregnancy may be necessary and has not been proven to be harmful to the fetus
- At delivery there may be more bleeding in the mother and transient irritability or sedation in the newborn

- Must weigh the risk of treatment (first trimester fetal development, third trimester newborn delivery) to the child against the risk of no treatment (recurrence of depression, maternal health, infant bonding) to the mother and child
- For many patients, this may mean continuing treatment during pregnancy
- Exposure to SSRIs early in pregnancy may be associated with increased risk of septal heart defects (absolute risk is small)
- SSRI use beyond the 20th week of pregnancy may be associated with increased risk of pulmonary hypertension in newborns, although this is not proven
- Exposure to SSRIs late in pregnancy may be associated with increased risk of gestational hypertension and preeclampsia
- Neonates exposed to SSRIs or SNRIs late in the third trimester have developed complications requiring prolonged hospitalization, respiratory support, and tube feeding; reported symptoms are consistent with either a direct toxic effect of SSRIs and SNRIs or, possibly, a drug discontinuation syndrome, and include respiratory distress, cyanosis, apnea, seizures, temperature instability, feeding difficulty, vomiting, hypoglycemia, hypotonia, hypertonia, hyperreflexia, tremor, jitteriness, irritability, and constant crying
- National Pregnancy Registry for Antidepressants: 1-866-961-2388, <https://womensmentalhealth.org/research/pregnancyregistry/antidepressants>

Breast Feeding

- Some drug is found in mother's breast milk
- Trace amounts may be present in nursing children whose mothers are on escitalopram
- If child becomes irritable or sedated, breast feeding or drug may need to be discontinued
- Immediate postpartum period is a high-risk time for depression, especially in women who have had prior depressive episodes, so drug may need to be reinstituted late in the third trimester or shortly after childbirth to prevent a recurrence during the postpartum period
- Must weigh benefits of breast feeding with risks and benefits of antidepressant treatment versus nontreatment to both the infant and the mother
- For many patients, this may mean continuing treatment during breast feeding

The art of psychopharmacology

Potential Advantages

- Patients taking concomitant medications (few drug interactions and fewer even than with citalopram)
- Patients requiring faster onset of action

Potential Disadvantages

- More expensive than citalopram in markets where citalopram is generic

Primary Target Symptoms

- Depressed mood
- Anxiety
- Panic attacks, avoidant behavior, reexperiencing, hyperarousal
- Sleep disturbance, both insomnia and hypersomnia

Pearls

- * May be among the best-tolerated antidepressants
- May have less sexual dysfunction than some other SSRIs
- May be better tolerated than citalopram
- Can cause cognitive and affective “flattening”
- * R-citalopram may interfere with the binding of S-citalopram at the serotonin transporter
- * For this reason, S-citalopram may be more than twice as potent as R,S-citalopram (i.e., citalopram)
- Thus, 10 mg starting dose of S-citalopram may have the therapeutic efficacy of 40 mg of R,S-citalopram
- Thus, escitalopram may have faster onset and better efficacy with reduced side effects compared to R,S-citalopram

- Some data may actually suggest remission rates comparable to SNRIs, but this is not proven
- * Escitalopram is commonly used with augmenting agents, as it is the SSRI with the least interaction at either CYP450 2D6 or 3A4, therefore causing fewer pharmacokinetically mediated drug interactions with augmenting agents than other SSRIs
- SSRIs may be less effective in women over 50, especially if they are not taking estrogen
- SSRIs may be useful for hot flushes in perimenopausal women
- Some postmenopausal women's depression will respond better to escitalopram plus estrogen augmentation than to escitalopram alone
- Nonresponse to escitalopram in elderly may require consideration of mild cognitive impairment or Alzheimer disease

Suggested Reading

Baldwin DS , Reines EH , Guiton C , Weiller E. Escitalopram therapy for major depression and anxiety disorders . *Ann Pharmacother* 2007 ;**41** (10):1583 –92.

.....

Bareggi SR , Mundo E , Dell'Oso B , Altamura AC . The use of escitalopram beyond major depression: pharmacological aspects, efficacy and tolerability in anxiety disorders . *Expert Opin Drug Metab Toxicol* 2007 ;**3** (5):741–53 .

.....
Burke WJ . Escitalopram . *Expert Opin Investig Drugs* 2002 ;**11** (10):1477
-86.

Esketamine

Therapeutics

Brands

- Spravato
- *see index for additional brand names*

Generic?

No

Class

- N-methyl-D-aspartate (NMDA) receptor antagonist

Commonly Prescribed for

(bold for FDA approved)

- **Treatment-resistant depression in adults (adjunct)**
- **Depressive symptoms in adults with major depressive disorder (MDD) with acute suicidal ideation or behavior (adjunct)**

How the Drug Works

- Esketamine is a nonselective, noncompetitive open channel inhibitor of the NMDA receptor; specifically, it binds to the phencyclidine site of the NMDA receptor

- This leads to downstream glutamate release and consequent stimulation of other glutamate receptors, including AMPA receptors
- Theoretically, esketamine may have antidepressant effects because activation of AMPA receptors leads to activation of signal transduction cascades, including mTORC1, and an increase in growth factors such as BDNF that cause the expression of synaptic proteins and an increase in the density of dendritic spines

How Long Until It Works

- Antidepressant effects can occur within 24 hours

If It Works

- Can immediately alleviate depressed mood and suicidal ideation

If It Doesn't Work

Try a traditional antidepressant or stimulation therapy

Best Augmenting Combos for Partial Response or Treatment Resistance

- Esketamine itself is an augmenting agent

Tests

Assess blood pressure prior to dosing: if baseline blood pressure is elevated (e.g., >140 mmHg systolic, >90 mmHg diastolic), consider the risks of short-term increases in blood pressure and benefit of

esketamine treatment in patients with treatment-resistant depression, and do not administer esketamine if an increase in blood pressure or intracranial pressure poses a serious risk

Given its potential to induce dissociative effects, carefully assess patients with psychosis before administering esketamine

Side effects

How Drug Causes Side Effects

- Direct effect on NMDA receptors

Notable Side Effects

Dissociation, dizziness, sedation, vertigo, anxiety, lethargy, feeling drunk, numbness, hypoesthesia

Nausea, vomiting

Lower urinary tract symptoms, including increased frequency of urination (pollakiuria)



Life-Threatening or Dangerous Side Effects

- Dissociation, sedation (requires monitoring for at least 2 hours at each treatment session)
- Increased blood pressure (peak at approximately 40 minutes post-administration and lasts approximately 4 hours)

- Short-term cognitive impairment (generally resolves by 2 hours post-dose)

Weight Gain



- Reported but not expected

Sedation



- Many experience and/or can be significant in amount

What to Do About Side Effects

- Use lower dose (56 mg rather than 84 mg)
- For CNS side effects, discontinuation of nonessential centrally acting medications may help
- If blood pressure remains high, promptly seek assistance from practitioners experienced in blood pressure management
- Refer patients experiencing symptoms of a hypertensive crisis (e.g., chest pain, shortness of breath) or hypertensive encephalopathy (e.g., sudden severe headache, visual disturbances, seizures, diminished consciousness, or focal neurological deficits) immediately for emergency care

Best Augmenting Agents for Side Effects

Many side effects cannot be improved with an augmenting agent

Dosing and use

Usual Dosage Range

- Induction phase (weeks 1–4): administer twice per week; day 1 starting dose 56 mg; subsequent doses 56 mg or 84 mg
- Maintenance phase (weeks 5–8): administer once weekly; 56 mg or 84 mg
- Maintenance phase (weeks 9 and after): administer every 2 weeks or once weekly (use least frequent dosing to maintain remission/response); 56 mg or 84 mg

Dosage Forms

- 56 mg dose kit: unit-dose carton containing two 28 mg nasal spray devices
- 84 mg dose kit: unit-dose carton containing three 28 mg nasal spray devices

How to Dose

- Given in conjunction with an oral antidepressant
- Administered intranasally under the supervision of a healthcare provider

- Each nasal spray device delivers 2 sprays containing a total of 28 mg of esketamine
- To prevent loss of medication, do not prime the device before use
- Patient should recline head at about 45 degrees during administration to keep medication inside the nose
- Patient should blow nose before first device only
- After patient self-administers both sprays (one in each nostril), check that the indicator shows no green dots; if there is a green dot, have patient spray again into the second nostril
- Use 2 devices (for a 56 mg dose) or 3 devices (for a 84 mg dose) with a 5-minute rest period between each device
- Blood pressure must be assessed before administration; if blood pressure is elevated, weigh the risks vs. benefits of esketamine administration
- Reassess blood pressure 40 minutes post-administration and thereafter as clinically warranted
- Patients must be monitored for at least 2 hours at each treatment session, followed by an assessment to determine when the patient is clinically stable and ready to leave the healthcare facility; this is due to the risks of sedation, dissociation, and hypertension
- If blood pressure is decreasing and patient is clinically stable for at least 2 hours, the patient may be discharged; if not, continue to monitor

- Advise the patient not to engage in potentially hazardous activities, such as driving or operating machinery, until the next day after a restful sleep; patients will need to arrange transportation home



Dosing Tips

Nausea and vomiting are potential side effects, so advise patients to avoid food for at least 2 hours before administration and liquids for at least 30 minutes before administration

Nasal corticosteroid or nasal decongestant should not be used within an hour prior to administration of esketamine

If a patient misses treatment sessions and there is a worsening of symptoms, consider returning to the previous dosing schedule (i.e., every 2 weeks to once weekly, weekly to twice weekly)

Evidence of therapeutic benefit should be evaluated at the end of the induction phase to determine the need for continued treatment

Overdose

- With racemic ketamine, symptoms of overdose include restlessness, psychosis, hallucinations, stupor, respiratory depression

Long-Term Use

- Long-term cognitive effects of esketamine have not been evaluated beyond 1 year

- Long-term cognitive and memory impairment have been reported with repeated ketamine misuse or abuse

Habit Forming

- Esketamine is a Schedule III drug and is available only through a restricted program under a Risk Evaluation and Mitigation Strategy (REMS)

How to Stop

- Taper not necessary

Pharmacokinetics

- Time to maximum plasma concentration is approximately 20–40 minutes after the last administered nasal spray
- Mean terminal half-life 7–12 hours
- Metabolized primarily by CYP450 2B6 and CYP450 3A4, and to a lesser extent by CYP450 2C9 and 2C19
- No significant actions on CYP450 enzymes, although esketamine has modest induction effects on CYP450 2B6 and 3A4 in human hepatocytes



Drug Interactions

Little potential to affect metabolism of drugs cleared by CYP450 enzymes

Use with caution with other drugs that are NMDA antagonists (amantadine, memantine, dextromethorphan)

Esketamine may increase the effects of other sedatives, including benzodiazepines, barbiturates, opioids, anesthetics, and alcohol

Concomitant use with stimulants or monoamine oxidase inhibitors may increase blood pressure

Other Warnings/Precautions

Because of the risks of sedation and dissociation and of abuse and misuse, esketamine is only available through a restricted program under a REMS; healthcare settings and pharmacies must be certified in the REMS program, and pharmacies must only dispense esketamine to healthcare settings that are certified in the program

Esketamine can raise blood pressure; patients with cardiovascular and cerebrovascular conditions and risk factors may be at an increased risk of associated adverse effects

Blood pressure should be monitored for at least 2 hours post-administration of esketamine, and prompt medical care should be sought if blood pressure remains high; refer patients experiencing symptoms of a hypertensive crisis or hypertensive encephalopathy for immediate emergency care

In patients with a history of hypertensive encephalopathy, more intensive monitoring, including more frequent blood pressure and symptom assessment, is warranted because these patients are at higher

risk for developing encephalopathy with even small increases in blood pressure

Because of the risks of delayed or prolonged sedation and dissociation, patients must be monitored for at least 2 hours at each treatment session, followed by an assessment to determine when the patient is clinically stable and ready to leave the healthcare facility

Because of the risk of dissociation, carefully assess patients with psychosis before administering esketamine and only initiate treatment if the potential benefits outweigh the risks

Monitor patients for activation of suicidal ideation

Esketamine may impair attention, judgment, thinking, reaction speed, and motor skills

Esketamine may impair ability to drive and operate machinery; patients should not drive or operate machinery until the next day after a restful sleep

When treating children, carefully weigh the risks and benefits of pharmacological treatment against the risks and benefits of nontreatment with antidepressants and make sure to document this in the patient's chart

Distribute the brochures provided by the FDA and the drug companies

Whenever possible, warn patients and their caregivers about the possibility of activating side effects, and advise them to report such symptoms immediately

Monitor patients for activation of suicidal ideation, especially children and adolescents

Do Not Use

- If patient has aneurysmal vascular disease (including thoracic and abdominal aorta, intracranial and peripheral arterial vessels) or arteriovenous malformation
- If patient has intracerebral hemorrhage
- If there is a proven allergy to esketamine or ketamine

Special populations

Renal Impairment

- No dose adjustment necessary

Hepatic Impairment

- Patients with moderate impairment may need to be monitored for adverse reactions for a longer period of time
- Not recommended for use in patients with severe impairment

Cardiac Impairment

- Contraindicated in patients with aneurysmal vascular disease (including thoracic and abdominal aorta, intracranial and peripheral arterial vessels) or arteriovenous malformation

- Patients with cardiovascular and cerebrovascular conditions and risk factors may be at an increased risk of associated adverse effects

Elderly

- In clinical trials, no differences were observed in safety profile between patients aged 65 years and older and patients younger than 65 years
- In clinical trials, C_{max} and AUC values were higher in patients aged 65 years and older than in patients younger than 65 years



Children and Adolescents

- Safety and efficacy have not been established



Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- Not recommended for use during pregnancy
- In pregnant primates, use of NMDA receptor antagonists during the period of peak brain development increased neuronal apoptosis in

the developing brain of the fetuses

- In rabbits, skeletal malformations were observed when ketamine was administered intranasally at estimated esketamine exposures of 0.3 times the maximum recommended human dose
- In rats, delay in sensorimotor development in pups was observed when esketamine was administered intranasally at exposures similar to the maximum recommended human dose
- Because of the risk for fetal harm, consider pregnancy planning and prevention in females of reproductive potential
- National Pregnancy Registry for Antidepressants: 1-866-961-2388, <https://womensmentalhealth.org/research/pregnancyregistry/antidepressants>

Breast Feeding

- Some drug is found in mother's breast milk
- Animal studies have shown neurotoxicity in juvenile animals when NMDA antagonists were administered during a window of vulnerability that correlates with exposures in the third trimester through the first several months of life, but this window may extend out to approximately 3 years of age in humans
- Recommended either to discontinue drug or bottle feed

The art of psychopharmacology

Potential Advantages

- Severely treatment-resistant depression, suicidal ideation

Potential Disadvantages

- Only available through a restricted program under a REMS
- Must be administered in the presence of a healthcare professional
- Requires monitoring for at least 2 hours post-administration

Primary Target Symptoms

- Treatment-resistant depression

Pearls

Esketamine is the “s” enantiomer of ketamine

Esketamine is not approved as an anesthetic, and its safety or efficacy as an anesthetic has not been established

In clinical trials, treatment-resistant depression was defined as a DSM-5 diagnosis for major depressive disorder (MDD) in patients who have not responded adequately to at least two different antidepressants of adequate dose and duration in the current depressive episode

Several studies suggest rapid reduction of depressive symptoms in patients with MDD who have active suicidal ideation with intent

Studies of esketamine or IV ketamine administered during psychotherapy sessions suggest potential improvement of substance

and alcohol abuse

Suggested Reading

Canuso CM Canuso CM , Singh JB Singh JB , Fedgchin M Fedgchin M , et al. Efficacy and safety of esketamine for the rapid reduction of symptoms of depression and suicidality in patients at imminent risk for suicide: results of a double-blind, randomized, placebo-controlled study . *Am J Psychiatry* 2018 ;**175** (7):620 –30 .

.....

Daly EJ Daly EJ , Trivedi MH Trivedi MH , Janik A Janik A , et al. Efficacy of esketamine nasal spray plus oral antidepressant treatment for relapse prevention in patients with treatment-resistant depression: a randomized clinical trial . *JAMA Psychiatry* 2019 ;**76** :893 –903 . doi: 10.1001/jamapsychiatry.2019.1189 .

.....

Fedgchin M Fedgchin M , Trivedi M Trivedi M , Daly E Daly E , et al. Efficacy and safety of fixed-dose esketamine nasal spray combined with a new oral antidepressant in treatment-resistant depression: results of a randomized, double-blind, active-controlled study (TRANSFORM-1) . *Int J Neuropsychopharmacol* 2019 ;**22** (10):616 –30 .

.....

Ochs-Ross R Ochs-Ross R , Daly EJ Daly EJ , Zhang Y Zhang Y , et al. Efficacy and safety of esketamine nasal spray plus an oral antidepressant in elderly patients with treatment-resistant depression –TRANSFORM-3 . *Am J Geriatr Psychiatry* 2019 ;**28** (2):121 –41 .

.....

Popova V Popova V , Daly EJ Daly EJ , Trivedi M Trivedi M , et al.
Efficacy and safety of flexibly dose esketamine nasal spray combined with
a newly initiated oral antidepressant in treatment-resistant depression: a
randomized double-blind active-controlled study . *Am J Psychiatry* 2019
;176 (6):428 –38 .

Estazolam

Therapeutics

Brands

- ProSom
- *see index for additional brand names*

Generic?

Yes

Class

- Neuroscience-based Nomenclature: GABA positive allosteric modulator (GABA-PAM)
- Benzodiazepine (hypnotic)

Commonly Prescribed for

(bold for FDA approved)

- **Insomnia characterized by difficulty in falling asleep, frequent nocturnal awakenings, and/or early morning awakenings**
- Catatonia

How the Drug Works

- Binds to benzodiazepine receptors at the GABA-A ligand-gated chloride channel complex
- Enhances the inhibitory effects of GABA
- Boosts chloride conductance through GABA-regulated channels
- Inhibitory actions in sleep centers may provide sedative hypnotic effects

How Long Until It Works

- Generally takes effect in less than an hour

If It Works

- Improves quality of sleep
- Effects on total wake-time and number of nighttime awakenings may be decreased over time

If It Doesn't Work

- If insomnia does not improve after 7–10 days, it may be a manifestation of a primary psychiatric or physical illness such as obstructive sleep apnea or restless leg syndrome, which requires independent evaluation
- Increase the dose
- Improve sleep hygiene
- Switch to another agent

Best Augmenting Combos for Partial Response or Treatment Resistance

- Generally, best to switch to another agent
- Trazodone
- Agents with antihistamine actions (e.g., diphenhydramine, TCAs)

Tests

- In patients with seizure disorders, concomitant medical illness, and/or those with multiple concomitant long-term medications, periodic liver tests and blood counts may be prudent

Side effects

How Drug Causes Side Effects

- Same mechanism for side effects as for therapeutic effects –namely due to excessive actions at benzodiazepine receptors
- Actions at benzodiazepine receptors that carry over to the next day can cause daytime sedation, amnesia, and ataxia
- Long-term adaptations in benzodiazepine receptors may explain the development of dependence, tolerance, and withdrawal

Notable Side Effects

- * Sedation, fatigue, depression
- * Dizziness, ataxia, slurred speech, weakness